

CSCE 411: Mastery Quiz 3

Name: _____

UIN: _____

Let

$\text{ONE-MORE} := \{ \langle P_1, P_2 \rangle \mid P_1 \text{ and } P_2 \text{ are algorithms such that } L(P_2) \subset L(P_1) \text{ and } |L(P_1) \setminus L(P_2)| = 1 \}$

(i.e. ONE-MORE is the language of (string encodings of) pairs of algorithms P_1, P_2 where the language of P_1 contains all the strings in the language of P_2 plus exactly one more string).

Prove that ONE-MORE is undecidable.

Solution: To prove that ONE-MORE is undecidable, we show that $\text{HALTS} \leq_T \text{ONE-MORE}$. Let ONEMOREDECIDER be a decider for ONE-MORE. We create the following decider HALTSDECIDER for HALTS:

$\text{HALTSDECIDER}(P, x)$

Write out the source code of the following algorithm HELPER1 :

$\text{HELPER1}(y)$

Simulate $P(x)$

If $y = \varepsilon$:

Return True

Return False

Write out the source code of the following algorithm HELPER2 :

$\text{HELPER2}(y)$

Return False

Return $\text{ONEMOREDECIDER}(\text{HELPER1}, \text{HELPER2})$

Correctness: We note that since HELPER2 returns False on all inputs, $L(\text{HELPER2}) = \emptyset$, whether or not $P(x)$ halts.

- Suppose $\langle P, x \rangle \in \text{HALTS}$. Then $P(x)$ halts, so $L(\text{HELPER1}) = \{\varepsilon\}$, and thus $|L(\text{HELPER1}) \setminus L(\text{HELPER2})| = |\{\varepsilon\}| = 1$. Thus $\langle \text{HELPER1}, \text{HELPER2} \rangle \in \text{ONE-MORE}$, so $\text{ONEMOREDECIDER}(\text{HELPER1}, \text{HELPER2})$ will return True, so $\text{HALTSDECIDER}(P, x)$ will return True.
- Suppose $\langle P, x \rangle \notin \text{HALTS}$. Then $P(x)$ loops, so $L(\text{HELPER1}) = \emptyset$, and thus $|L(\text{HELPER1}) \setminus L(\text{HELPER2})| = |\emptyset| = 0 \neq 1$. Therefore $\langle \text{HELPER1}, \text{HELPER2} \rangle \notin \text{ONE-MORE}$, so $\text{ONEMOREDECIDER}(\text{HELPER1}, \text{HELPER2})$ will return False, so $\text{HALTSDECIDER}(P, x)$ will return False.